

1. Project Overview

The **Flask Authors' API** is a structured system designed to streamline book management for authors, publishing companies, and administrators. It provides secure authentication and **role-based access control**, ensuring that only authorized users can create, update, or delete books. The API also facilitates structured relationships between books, authors, and publishing companies, making it easier to organize and retrieve book-related information efficiently.

This API addresses key challenges such as **unauthorized book modifications, inefficient book tracking, and security concerns**. By implementing **JWT-based authentication**, it ensures that only registered users can access the system. Additionally, the API enforces **proper access control**, preventing unauthorized users from editing or deleting books they do not own. Publishing companies can oversee the books associated with their brand, while admins maintain system integrity by managing users and content.

The API's key features include **user authentication, book management, company management, and user role control**. It provides a **scalable solution** for authors to manage their work, publishers to organize their collections, and readers to explore available books. By offering a structured and secure platform, this API enhances efficiency in book management and publishing workflows.

2. Project Setup

Requirements

- alembic==1.14.0
- bcrypt==4.2.1
- blinker==1.9.0
- click==8.1.8
- colorama==0.4.6
- Flask==3.1.0
- Flask-Bcrypt==1.0.1
- Flask-JWT-Extended==4.7.1
- Flask-Migrate==4.0.7
- Flask-SQLAlchemy==3.1.1
- greenlet==3.1.1
- itsdangerous==2.2.0
- Jinja2==3.1.5
- Mako==1.3.8
- MarkupSafe==3.0.2
- PyJWT==2.10.1

- PyMySQL==1.1.1
- python-dotenv==1.0.1
- SQLAlchemy==2.0.36
- typing_extensions==4.12.2
- validators==0.34.0
- Werkzeug==3.1.3
- XAMPP (MySQL Database)

Installation

Clone the repository

```
git clone https://github.com/yourusername/project.git
```

Navigate to the project folder

```
cd project
```

Create a virtual environment

```
python -m venv venv
```

```
source venv/bin/activate # For Mac/Linux
```

```
venv\Scripts\activate # For Windows
```

Install dependencies

```
pip install -r requirements.txt
```

Set up environment variables

```
cp .env.example .env
```

Database Setup

Create the database

```
flask db init
```

```
flask db migrate -m "Initial migration"
```

```
flask db upgrade
```

Running the Application

```
flask run
```

3. API Documentation

Postman for documentation

Authentication

Register User

POST /auth/api/v1/auth/register

Request Body:

```
{  
  "first_name": "Ramula",  
  "last_name": "rubuga",  
  "email": "rubugaramula@gmail.com",  
  "contact": "0755784524",  
  "password": "ramula25634",  
  "biography": "A Novelist",  
  "user_type": "author"  
}
```

Response:

```
{  
  "message": "User registered successfully"  
}
```

Login

POST /auth/login

Response:

```
{  
  "message": "Ramula rubuga has been successfully created as an author",  
  "user": {  
    "biography": "A Novelist",  
    "contact": "0755784524",  
    "created_at": "Wed, 02 Apr 2025 19:27:43 GMT",  
    "email": "rubugaramula@gmail.com",  
    "first_name": "Ramula",  
    "id": 9,  
    "last_name": "rubuga",  
    "type": "author"  
  }  
}
```

Login User

POST/api/v1/auth/login

Request Body

```
{  
  
  "email": "rubugaramula@gmail.com",  
  "password": "ramula25634"  
  
}
```

Response

```
"message": "You have successfully logged into your account.",

"user": {

  "access_token":
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJmcmVzaCI6ZmFsc2UsImhhdCI6MTc0MzYxMjg4MiwianRpljoiMjNmYTQxZWVtYWQ3NC00M2ExLThjOTQ5ZGI1NTJlOTU4IiwidHlwZSI6ImFjY2VzcyIsInN1YiI6IjkiLCJuYmYiOiJlY3NDM2MTI4ODIsImNzcmYiOiJmOTlkNzI4YS0yMDA5LTRlNGYtOTBhMC0xMmYzYTk3ZjZk5ZTMiLCJleHAiOiJlY3NDM2MTM3ODJ9.6ZkgH-FOV0yPUmmFK_u8nGUwoLpqwR3lL-KoHd9W88g",

  "email": "rubugaramula@gmail.com",

  "id": 9,

  "refresh_token":
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJmcmVzaCI6ZmFsc2UsImhhdCI6MTc0MzYxMjg4MiwianRpljoiMThmNTE2ZDctYTE1Yi00YWM0LThiZWVtN2lwMjQwYzZjZTM0IiwidHlwZSI6ImFjY2VzcyIsInN1YiI6IjkiLCJuYmYiOiJlY3NDM2MTI4ODIsImNzcmYiOiJmOTlkNzI4YS0yMDA5LTRlNGYtOTBhMC0xMmYzYTk3ZjZk5ZTMiLCJleHAiOiJlY3NDM2MTM3ODJ9.6ZkgH-FOV0yPUmmFK_u8nGUwoLpqwR3lL-KoHd9W88g",

  "type": "author",

  "username": "Ramula rubuga"

}

}
```

Log Out

POST/api/v1/auth/logout

Request Body:

Bearer token of logged in user:

Example:

```
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJmcmVzaCI6ZmFsc2UsImhhdCI6MTc0MzYxMjg4MiwianRpljoiNTUxMzliYTQxZWVtYWQ3NC00M2ExLThjOTQ5ZGI1NTJlOTU4IiwidHlwZSI6ImFjY2VzcyIsInN1YiI6IjkiLCJuYmYiOiJlY3NDM2MTI4ODIsImNzcmYiOiJmOTlkNzI4YS0yMDA5LTRlNGYtOTBhMC0xMmYzYTk3ZjZk5ZTMiLCJleHAiOiJlY3NDM2MTM3ODJ9.6ZkgH-FOV0yPUmmFK_u8nGUwoLpqwR3lL-KoHd9W88g
```

Response

```
{
  "message": "You have successfully logged out."
}
```

Users API

Get all users

GET/api/v1/users/

Request

Bearer token of currently logged in user

Example:

eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJmcmVzaCI6ZmFsc2UsImhhbmciOiJhbmM0MzY1ODk1OCwianRpIjoibG9YTc0NzQtYTNTkNyO0MGEzLWVzNWltZTc5MmM0NzlxMDhiliwidHlwZSI6ImFjY2VzcyIsInN1YiI6IjliLCJuYmYiOiJlY3NDM2NTg5NTg5ImNzcmYiOiJhbmMzZG1ZWQyYS0zY2RhLTQ1OGYtYjQ1YS00MzFmMDY4YmI1NTMiLCJleHAiOiJlY3NDM2NTk4NTh9.iqxI0JHz107gs3aECtljKJe7nz9f2q470VPhIQVhW5A

Response

```
{
  "message": "All users successfully retrieved.",
  "total_users": 9,
  "users": [
    {
      "contact": "0773874765",
      "created_at": "Fri, 03 Jan 2025 15:55:28 GMT",
      "email": "phionaaladina@gmail.com",
      "first_name": "Fiona",
      "id": 1,
      "last_name": "Aladina",
      "type": "author",
      "username": "Fiona Aladina"
    }
  ]
}
```

},

{

"contact": "0773874755",

"created_at": "Fri, 03 Jan 2025 16:10:01 GMT",

"email": "admin@gmail.com",

"first_name": "Admin",

"id": 2,

"last_name": "System",

"type": "admin",

"username": "Admin System"

},

{

"contact": "076784956",

"created_at": "Wed, 05 Mar 2025 09:41:20 GMT",

"email": "leila@gmail.com",

"first_name": "Leile",

"id": 3,

"last_name": "Ashati",

"type": "author",

"username": "Leile Ashati"

},

{

"contact": "0767849100",

"created_at": "Wed, 12 Mar 2025 15:03:59 GMT",

"email": "gorret@gmail.com",

"first_name": "Tee",

"id": 4,

"last_name": "Ashati",

```
"type": "author",
"username": "Tee Ashati"
},
{
  "contact": "076700000",
  "created_at": "Thu, 13 Mar 2025 09:38:51 GMT",
  "email": "patrickstar@gmail.com",
  "first_name": "Patrick",
  "id": 5,
  "last_name": "Star",
  "type": "author",
  "username": "Patrick Star"
},
{
  "contact": "076701100",
  "created_at": "Sat, 15 Mar 2025 17:46:22 GMT",
  "email": "spongebob@gmail.com",
  "first_name": "Spongebob",
  "id": 6,
  "last_name": "Squarepants",
  "type": "author",
  "username": "Spongebob Squarepants"
},
{
  "contact": "0767012222",
  "created_at": "Wed, 02 Apr 2025 17:10:33 GMT",
  "email": "arindasherinah1@gmail.com",
  "first_name": "Sherinah",
```



```
"id": 7,
"last_name": "Arinda",
"type": "author",
"username": "Sherinah Arinda"
},
{
  "contact": "0767012220",
  "created_at": "Wed, 02 Apr 2025 17:10:33 GMT",
  "email": "test@gmail.com",
  "first_name": "test",
  "id": 8,
  "last_name": "user",
  "type": "author",
  "username": "test user"
},
{
  "contact": "0755784524",
  "created_at": "Wed, 02 Apr 2025 19:27:43 GMT",
  "email": "rubugaramula@gmail.com",
  "first_name": "Ramula",
  "id": 9,
  "last_name": "rubuga",
  "type": "author",
  "username": "Ramula rubuga"
}
]
}
```

Get all authors

GET/api/v1/users/authors

Request

Bearer token of currently logged in user

Example:

eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJmcmVzaCI6ZmFsc2UsImhdCI6MTc0MzY1ODk1OCwianRpIjoibjg0YTc0NzQtYTlnNy00MGExLWEzNWltZTc5MmM0NzlxMDhiliwidHlwZSI6ImFjY2VzcylsbnN1Yil6IjiiLCJuYmYiOiJlE3NDM2NTg5NTgsImNzcmYiOiJhMzg1ZWQyYS0zY2RhLTQ1OGYtYjQ1YS00MzFmMDY4Yml1NTMiLCJleHAiOiJlE3NDM2NTk4NTh9.iqxI0JHz107gs3aECtljKJe7nz9f2q470VPhIQVhW5A

Response

```
{
  "authors": [
    {
      "biography": "",
      "books": [],
      "companies": [],
      "contact": "0773874765",
      "created_at": "Fri, 03 Jan 2025 15:55:28 GMT",
      "email": "phionaaladina@gmail.com",
      "first_name": "Fiona",
      "id": 1,
      "last_name": "Aladina",
      "username": "Fiona Aladina"
    },
    {
      "biography": "New York Best times seller",
      "books": [],
```

```
"companies": [  
  {  
    "id": 3,  
    "name": "WITI lib",  
    "origin": "Uganda"  
  },  
  {  
    "id": 4,  
    "name": "WITI lissb",  
    "origin": "Uganda"  
  },  
  {  
    "id": 5,  
    "name": "WITI liss ghghb",  
    "origin": "Uganda"  
  },  
  {  
    "id": 7,  
    "name": "WIT Net",  
    "origin": "Uganda"  
  }  
],  
"contact": "076784956",  
"created_at": "Wed, 05 Mar 2025 09:41:20 GMT",  
"email": "leila@gmail.com",  
"first_name": "Leile",  
"id": 3,  
"last_name": "Ashati",
```

```
"username": "Leile Ashati"
},
{
  "biography": "New York Best times seller",
  "books": [],
  "companies": [
    {
      "id": 8,
      "name": "STATIONARY",
      "origin": "Uganda"
    }
  ],
  "contact": "0767849100",
  "created_at": "Wed, 12 Mar 2025 15:03:59 GMT",
  "email": "gorret@gmail.com",
  "first_name": "Tee",
  "id": 4,
  "last_name": "Ashati",
  "username": "Tee Ashati"
},
{
  "biography": "New York Best times seller",
  "books": [],
  "companies": [],
  "contact": "076700000",
  "created_at": "Thu, 13 Mar 2025 09:38:51 GMT",
  "email": "patrickstar@gmail.com",
  "first_name": "Patrick",
```

```
"id": 5,

"last_name": "Star",

"username": "Patrick Star"

},

{

  "biography": "New York Best times seller",

  "books": [

    {

      "created_at": "Sat, 15 Mar 2025 20:25:51 GMT",

      "description": "Exciting Book for Adventurers",

      "genre": "Fiction",

      "id": 1,

      "image": "https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcSlvweiJVyPKy79xaSJPT9oq6dPQH6OcxfpjA&s"

    },

    {

      "price": 25000,

      "price_unit": "UGX",

      "publication_date": "0000-00-00",

      "title": "Spongebob Squarepants' Adventures"

    }

  ],

  "companies": [

    {

      "id": 9,

      "name": "SpongeBob Chronicles",

      "origin": "US"

    }

  ],

  "contact": "076701100",
```

```
"created_at": "Sat, 15 Mar 2025 17:46:22 GMT",
"email": "spongebob@gmail.com",
"first_name": "Spongebob",
"id": 6,
"last_name": "Squarepants",
"username": "Spongebob Squarepants"
},
{
  "biography": "A dedicated author",
  "books": [
    {
      "created_at": "Wed, 02 Apr 2025 17:10:33 GMT",
      "description": "For basic skills",
      "genre": "Fiction",
      "id": 4,
      "image": "https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcSlvweiJVyPKy79xaSJPT9oq6dPQH6OcxfpjA&s"
    },
    {
      "price": 60000,
      "price_unit": "UGX",
      "publication_date": "0000-00-00",
      "title": "Life skills"
    }
  ],
  "companies": [],
  "contact": "0767012222",
  "created_at": "Wed, 02 Apr 2025 17:10:33 GMT",
  "email": "arindasherinah1@gmail.com",
  "first_name": "Sherinah",
```

```
"id": 7,
"last_name": "Arinda",
"username": "Sherinah Arinda"
},
{
  "biography": "A Novelist",
  "books": [],
  "companies": [
    {
      "id": 10,
      "name": "witiwe",
      "origin": "Uganda"
    }
  ],
  "contact": "0767012220",
  "created_at": "Wed, 02 Apr 2025 17:10:33 GMT",
  "email": "test@gmail.com",
  "first_name": "test",
  "id": 8,
  "last_name": "user",
  "username": "test user"
},
{
  "biography": "A Novelist",
  "books": [
    {
      "created_at": "Wed, 02 Apr 2025 19:27:43 GMT",
      "description": "All",
```

```
    "genre": "Animation",
    "id": 6,
    "image": "https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcSlvweiJVyPKy79xaSJPT9oq6dPQH6OcxfpjA&s"
,
    "price": 10000,
    "price_unit": "UGX",
    "publication_date": "0000-00-00",
    "title": "Believe"
  }
],
"companies": [
  {
    "id": 11,
    "name": "Rue",
    "origin": "kenya"
  }
],
"contact": "0755784524",
"created_at": "Wed, 02 Apr 2025 19:27:43 GMT",
"email": "rubugaramula@gmail.com",
"first_name": "Ramula",
"id": 9,
"last_name": "rubuga",
"username": "Ramula rubuga"
}
],
"message": "All authors successfully retrieved.",
"total_authors": 8
```



```
}
```

Get author by ID

GET /api/v1/users/author/{id}

Request

Bearer token of currently logged in user

Example:

```
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJmcmVzaCI6ZmFsc2UsImIhdCI6MTc0MzY1ODk1OCwianRpIjojNjg0YTc0NzQtYTlkNy00MGEzLWEzNWltZTc5MmM0NzIxMDhiliwidHlwZSI6ImFjY2VzcyIsInN1Yil6IjliLCJuYmYiOiJlY3NDMDNTg5NTgsImNzcmYiOiJhMzg1ZWQyYS0zY2RhLTQ1OGYtYjQ1YS00MzFmMDY4YmI1NTMiLCJleHAiOiJlY3NDMDNTk4NTh9.iqxI0JHz107gs3aECtljKJe7nz9f2q470VPhIQVhW5A
```

Response

```
"author": {
  "biography": "New York Best times seller",
  "books": [],
  "companies": [],
  "contact": "076700000",
  "created_at": "Thu, 13 Mar 2025 09:38:51 GMT",
  "email": "patrickstar@gmail.com",
  "first_name": "Patrick",
  "id": 5,
  "last_name": "Star",
  "username": "Patrick Star"
},
"message": "Author details successfully retrieved."
}
```

Companies API

Create a company

POST/api/v1/companies/create

Request Body

```
{  
  "name": "Rue",  
  "origin": "kenya",  
  "description": "Believe"  
}
```

Response

```
{  
  "company": {  
    "description": "Believe",  
    "id": 11,  
    "name": "Rue",  
    "origin": "kenya"  
  },  
  "message": "Rue has been successfully created as a company by user {user_id}"  
}
```

Get All Books

GET /api/v1/books/

Request

Bearer token of logged in user:

eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJmcmVzaCI6ZmFsc2UsImhhbmQiOi6MTc0MzYxMjEwMywianRpIjoibWZlbiYTYtMjVkNS00ZTE2LTgzMjgtZTY5ZjBhYmYyYjA4IiwidHlwZSI6ImFjcyIsbnN1Yil6IjkiLCJuYmYiOiJlE3NDM2MTIxMDMsImNzcmYiOiIyNDgwYzdjMy03NmM1LTQzYjYtYmRkNS00NTVkMDBjNTI2YWWEiLCJleHAiOiJlE3NDM2MTMwMDN9.VJMk1InxnCn8EqO6-DDvPgkUkg8LHhWTr1M7F3aLOXY

```
"books": [
  {
    "company": {
      "created_at": "Sat, 15 Mar 2025 17:54:03 GMT",
      "description": "The Best Stories Collection",
      "id": 9,
      "name": "SpongeBob Chronicles",
      "origin": "US"
    },
    "created_at": "Sat, 15 Mar 2025 20:25:51 GMT",
    "description": "Exciting Book for Adventurers",
    "genre": "Fiction",
    "id": 1,
    "image": "https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcSlvweiJVyPKy79xaSJPT9oq6dPQH6OcxfpjA&s"
    ,
    "isbn": "BOOK111-02-2024",
    "pages": 350,
    "price": 25000,
    "price_unit": "UGX",
    "publication_date": "0000-00-00",
    "title": "Spongebob Squarepants' Adventures"
```

```

},
{
  "company": {
    "created_at": "Wed, 12 Mar 2025 16:34:48 GMT",
    "description": "OneStop Place for Children's Books",
    "id": 7,
    "name": "WIT Net",
    "origin": "Uganda"
  },
  "created_at": "Wed, 02 Apr 2025 17:10:33 GMT",
  "description": "For basic skills",
  "genre": "Fiction",
  "id": 4,
  "image": "https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcSlvweiJVyPKy79xaSJPT9oq6dPQH6OcxfpjA&s"
,
  "isbn": "BOOK111-022-202300",
  "pages": 400,
  "price": 40000,
  "price_unit": "UGX",
  "publication_date": "0000-00-00",
  "title": "Life skills"
},
{
  "company": {
    "created_at": "Wed, 02 Apr 2025 17:10:33 GMT",
    "description": "Here for you",
    "id": 10,
    "name": "witiwe",

```

```
    "origin": "Uganda"
  },
  "created_at": "Wed, 02 Apr 2025 17:10:33 GMT",
  "description": "For",
  "genre": "Fiction",
  "id": 5,
  "image": "https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcSlvweiJVyPKy79xaSJPT9oq6dPQH6OcxfpjA&s"
,
  "isbn": "BOOK111-02-202300",
  "pages": 400,
  "price": 5000,
  "price_unit": "UGX",
  "publication_date": "0000-00-00",
  "title": "Yap"
},
{
  "company": {
    "created_at": "Wed, 02 Apr 2025 19:27:43 GMT",
    "description": "Believe",
    "id": 11,
    "name": "Rue",
    "origin": "kenya"
  },
  "created_at": "Wed, 02 Apr 2025 19:27:43 GMT",
  "description": "All",
  "genre": "Animation",
  "id": 6,
```

```
{
  "book": {
    "company": {
```

```
    "created_at": "Sat, 15 Mar 2025 17:54:03 GMT",
    "description": "The Best Stories Collection",
    "id": 9,
    "name": "SpongeBob Chronicles",
    "origin": "US"
  },
  "created_at": "Sat, 15 Mar 2025 20:25:51 GMT",
  "description": "Exciting Book for Adventurers",
  "genre": "Fiction",
  "id": 1,
  "image": "https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcSlvweiJVyPKy79xaSJPT9oq6dPQH6OcxfpjA&s"
,
  "isbn": "BOOK111-02-2024",
  "pages": 350,
  "price": 25000,
  "price_unit": "UGX",
  "publication_date": "0000-00-00",
  "title": "Spongebob Squarepants' Adventures"
},
"message": "Book details successfully retrieved."
}
```

Update Book

PATCH / /api/v1/books/edit/{id}

Request Body:

```
{
```

```
"price": 60000
}
```

Delete Book

DELETE / api/v1/books/delete/{id}

Request

Bearer token of currently logged in user

Response

```
{
  "message": "Book has been successfully deleted"
}
```

4. Project Structure

/Flask_Authors_API/

```
|— app/
|   |— models.py
|   |— controllers/
|   |   |— auth.py
|   |   |— books.py
|   |   |— users.py
|   |   |— companies.py
|   |— routes.py
|   |— config.py
|
|   |— controllers/
|   |   |— Auth/
```



```
| | | ├── __pycache__
| | | ├── auth_controller.py
| | | ├── __init__.py
| | ├── Books/
| | | ├── __pycache__
| | | ├── book_controller.py
| | | ├── __init__.py
| | ├── Companies/
| | | ├── __pycache__
| | | ├── companies_controller.py
| | | ├── __init__.py
| | ├── Users/
| | | ├── __pycache__
| | | ├── user_controller.py
| | | ├── __init__.py
| | ├── __pycache__
| | ├── __init__.py
| ├── models\
| ├── __pycache__
| ├── extensions.py
| ├── status_codes.py
| ├── __init__.py
|── authors_venv
|── migrations
|── __pycache__
|── .env
|── .gitignore
```

|— config.py
|— requirements.txt
|— run.py

5. Roles & Permissions

Role	Permissions
Admin	Add/Edit/Delete any company or book
User	Can add/view a company or book

6. Contribution Guide

If it's an open-source project:

1. Fork the repo
2. Create a new branch
3. Commit changes
4. Open a pull request